

Rui Paulo da Silva MARTINS

professor

基本信息

姓名: Rui Paulo da Silva MARTINS 所属院校: 澳门大学

地区: China-Macau

荣誉称号

★ IEEE Fellow

这些成就代表了在学术领域的最高认可，展现了卓越的研究能力和影响力。

学科领域

微电子科学与工程

集成电路设计与集成系统

研究领域

射频集成电路

模拟与数模混合电路

低功耗电路与能量管理

混合信号集成技术

微电子器件与制造

模拟与射频IC设计

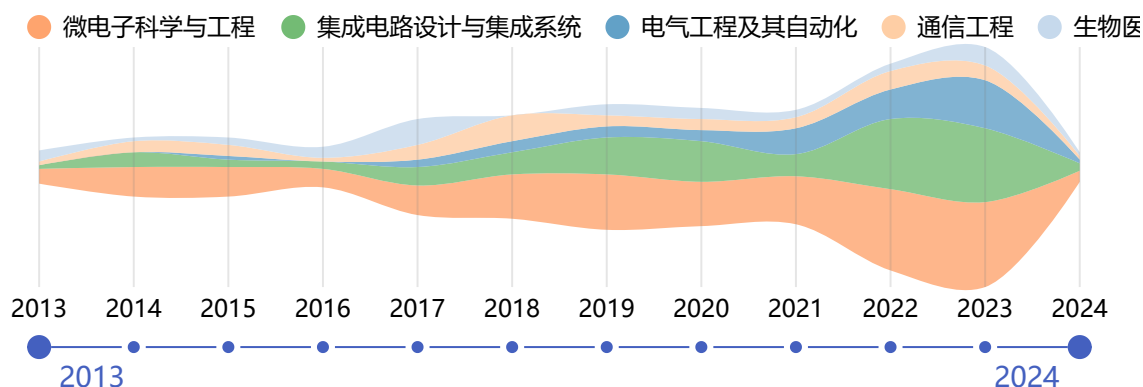
超大规模集成电路设计

电力电子与电机驱动

个人简介

Rui Paulo da Silva MARTINS, also known as 馬許願, is a distinguished Chair Professor and Vice-Rector (Global Affairs) at his institution. He also serves as the Director of the Institute of Microelectronics. A Fellow of IEEE (USA) and a Full Member of the Academy of Sciences of Lisbon, Portugal, his research focuses on Electronics, particularly Analog and Mixed-Signal VLSI.

近年发表的论文



A Radio-Frequency Cross-Connected Rectifier With LC Source Degeneration

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Chen, Qiuji; Huang, Mo; Xia, Tian; Yin, Jun; Liu, Haiwen; Martins, Rui P.; Lu, Yan

被引用次数: 0

2024

Absolute quantification of nucleic acid on digital microfluidics platform based on superhydrophobic-superhydrophilic micropatterning

SENSORS AND ACTUATORS B-CHEMICAL

Meng, Li; Li, Mingzhong; Xu, Zhenyu; Lv, Aman; Jia, Yanwei; Chen, Meiwan; Mak, Pui-In; Martins, Rui P.; Law, Man-Kay

被引用次数: 1

2024

An ELDC-Free 4th-Order CT SDM Facilitated by 2nd-Order NS CT-SAR and AC-Coupled Negative-R

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Xu, Zixuan; Xing, Kai; Zhu, Yan; Martins, Rui P.; Chan, Chi-Hang

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2024

A Fully Synthesizable All-Digital Dual-Loop Distributed Low-Dropout Regulator

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Mao, Xiangyu; Lu, Yan; Martins, Rui P.

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2024

A 93.4% Peak Efficiency CLOAD-Free Multi-Phase Switched-Capacitor DC-DC Converter Achieving a Fast DVS up to 222.5 mV/ns

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Li, Feiyu; Fang, Qishen; Wu, Jiangchao; Jiang, Yang; Mak, Pui-In; Martins, Rui P.; Law, Man-Kay

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2024

A 12-/13.56-MHz Crystal Oscillator With Binary-Search-Assisted Two-Step Injection Achieving 5.0-nJ Startup Energy and 45.8-μs Startup Time

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Li, Haihua; Lei, Ka-Meng; Martins, Rui P.; Mak, Pui-In

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2024

A syndromic diagnostic assay on a macrochannel-to-digital microfluidic platform for automatic identification of multiple respiratory pathogens

LAB ON A CHIP

Dong, Cheng; Li, Fei; Sun, Yun; Long, Dongling; Chen, Chunzhao; Li, Mengyan; Wei, Tao; Martins, Rui P.; Chen, Tianlan; Mak, Pui-In

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2023

A 3.6-GHz Type-II Sampling PLL With a Differential Parallel-Series Double-Edge S-PD Scoring 43.1-fsRMSJitter,-258.7-dB FOM, and -75.17-dBc Reference Spur

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Huang, Yunbo; Chen, Yong; Zhao, Bo; Mak, Pui-In; Martins, Rui P.

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2023

A 3.78-GHz Type-I Sampling PLL With a Fully Passive KPD-Doubled Master-Slave S-PD Measuring 39.6-fsRMS Jitter,-260.2-dB FOM, and -70.96-dBc Reference Spur

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

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2023

Nucleic acid analysis on electrowetting-based digital microfluidics

TRAC-TRENDS IN ANALYTICAL CHEMISTRY

Shen, Ren; Lv, A'man; Yi, Shuhong; Wang, Ping; Mak, Pui-In; Martins, Rui P.; Jia, Yanwei

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2023

A 6.78-MHz Wireless Power Transfer System With Inherent Wireless Phase Shift Control Without Feedback Data Sensing Coil

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Bai, Xianglong; Lu, Yan; Zhan, Chenchang; Martins, Rui P.

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2023

A 3.07 mW 30 MHz-BW 73.2 dB-SNDR Time-Interleaved Noise-Shaping SAR ADC With Self-Coupling Second-Order Error-Feedforward

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Zhao, Shulin; Guo, Mingqiang; Qi, Liang; Xu, Dengke; Wang, Guoxing; Martins, Rui P.; Sin, Sai-Weng

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Modeling-Attack-Resistant Strong PUF Exploiting Stagewise Obfuscated Interconnections With Improved Reliability

IEEE INTERNET OF THINGS JOURNAL

Xu, Chongyao; Zhang, Litao; Law, Man-Kay; Zhao, Xiaojin; Mak, Pui-In; Martins, Rui P.

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pH Regulator on Digital Microfluidics with Pico-Dosing Technique

BIOSENSORS-BASEL

Li, Haoran; Peng, Tao; Zhong, Yunlong; Liu, Meiqing; Mak, Pui-In; Martins, Rui P.; Wang, Ping; Jia, Yanwei

A 95% Peak Efficiency Modified KY Converter With Improved Flying Capacitor Charging in DCM for IoT Applications

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Pan, Caolei; Zeng, Wen-Liang; Lam, Chi-Seng; Sin, Sai-Weng; Zhan, Chenchang; Martins, Rui P.

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A Continuous-Output-Current Buck-Boost Converter Without Right-Half-Plane-Zero (RHPZ)

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Pan, Caolei; Zhan, Chenchang; Martins, Rui P.; Lam, Chi-Seng

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A 1224 V-Input HV-LV-Separated Hybrid SC PoL Converter With 355 mW/mm³ Power Density at 3 A Load Current and 15.2 mm³ Power Passives

IEEE TRANSACTIONS ON POWER ELECTRONICS

Zhang, Xiongjie; Ma, Qiaobo; Zhao, Anyang; Jiang, Yang; Law, Man-Kay; Jiang, Junmin; Takamiya, Makoto; Martins, Rui P.; Mak, Pui-In

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2023

A Subthreshold Operation Series-Parallel Charge Pump Incorporating Dynamic Source-Fed Oscillator for Wide-Input-Voltage Energy Harvesting Application

IEEE ACCESS

Kee, Yong Jack; Ramiah, Harikrishnan; Churchill, Kishore Kumar Pakkirisami; Chong, Gabriel; Mekhilef, Saad; Lai, Nai Shyan; Chen, Yong; Mak, Pui-In; Martins, Rui P.

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Design Trends and Perspectives of Digital Low Dropout Voltage Regulators for Low Voltage Mobile Applications: A Review

IEEE ACCESS

Lai, Li Fang; Ramiah, Harikrishnan; Tan, Yee-Chyan; Lai, Nai Shyan; Lim, Chee-Cheow; Chen, Yong; Mak, Pui-In; Martins, Rui P.

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Design and Implementation of Hybrid DC-DC Converter: A Review

IEEE ACCESS

Chang, Teck Seong; Ramiah, Harikrishnan; Jiang, Yang; Lim, Chee Cheow; Lai, Nai Shyan; Mak, Pui-In; Martins, Rui P.

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High-Performance Multiband Ambient RF Energy Harvesting Front-End System for Sustainable IoT Applications-A Review

IEEE ACCESS

Lee, Yi Chen; Ramiah, Harikrishnan; Choo, Alexander; Churchill, Kishore Kumar Pakkirisami; Lai, Nai Shyan; Lim, Chee Cheow; Chen, Yong; Mak, Pui-In; Martins, Rui P.

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2023

A 52.5-dB 2x Time-Interleaved 2.8-GS/s SAR ADC With 5-bit/Cycle Time-Domain Quantization and a Compact Signal DAC

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2023

A Second-Order NS Pipelined SAR ADC With Quantization-Prediction-Unrolled Gain Error Shaping and Fully Passive Integrator

IEEE JOURNAL OF SOLID-STATE CIRCUITS

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A 47-nW Voice Activity Detector (VAD) Featuring a Short-Time CNN Feature Extractor and an RNN-Based Classifier With a Non-Volatile CAP-ROM

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lin, Jinhai; Un, Ka-Fai; Yu, Wei-Han; Martins, Rui P.; Mak, Pui-In

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A 12-to-1 V Quad-Output Switched-Capacitor Buck Converter With Shared DC Capacitors

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Hu, Tingxu; Huang, Mo; Martins, Rui P.; Lu, Yan

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2023

A 14b 500 MS/s Single-Channel Pipelined-SAR ADC With Reference Ripple Mitigation Techniques and Adaptively Biased Floating Inverter Amplifier

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Jiang, Wenning; Zhu, Yan; Chen, Chixiao; Xu, Hao; Liu, Qi; Liu, Ming; Martins, Rui P.; Chan, Chi-Hang

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All Rivers Flow to the Sea: A High-Density Wireless Power Receiver With Split-Dual-Path and Hybrid-Quad-Path Step-Down Rectifying Conversion

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lin, Zixiao; Lu, Yan; Mao, Fangyu; Wang, Chuang; Martins, Rui P.

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An SC-Parallel-Inductor Hybrid Buck Converter With Reduced Inductor Voltage and Current

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Cai, Guigang; Lu, Yan; Martins, Rui P.

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2023

A 0.4-V 0.0294-mm² Resistor-Based Temperature Sensor Achieving ± 0.24 °C p2p Inaccuracy From -40°C to 125°C and 385 fJ • K² Resolution FoM in 65-nm CMOS

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Shi, Dan; Lei, Ka-Meng; Martins, Rui P.; Mak, Pui-In

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A Miniaturized 3-D-MRI Scanner Featuring an HV-SOI ASIC and Achieving a 10 x 8 x 8 mm³ Field of View

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Fan, Shuhao; Zhou, Qi; Lei, Ka-Meng; Mak, Pui-In; Martins, Rui P.

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A 10.5 W, 93% Efficient Dual-Path Hybrid(DPH)-Based DC-DC Converter Incorporating a Continuous-Current-Input Switched-Capacitor Stage and Enhanced IL Reduction for 12 V/24 V Inputs

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Ma, Qiaobo; Zhang, Xiongjie; Zhao, Anyang; Li, Huihua; Jiang, Yang; Law, Man-Kay; Takamiya, Makoto; Martins, Rui P.; Mak, Pui-In

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Analysis and Design of a 15.2-to-18.2-GHz Inverse-Class-F VCO With a Balanced Dual-Core Topology Suppressing the Flicker Noise Upconversion

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A Level Shifter With Almost Full Immunity to Positive dv/dt for Buck Converters

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A Fully Integrated CMOS Tri-Band Ambient RF Energy Harvesting System for IoT Devices

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Yong, Jack Kee; Lian, Wen Xun; Ramiah, Harikrishnan; Churchill, Kishore Kumar Pakkirisami; Chong, Gabriel; Lai, Nai Shyan; Chen, Yong; Mak, Pui-In; Martins, Rui P.

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An 1 V Supply, 740 nW, 8.7 ppm/°C Bandgap Voltage Reference With Segmented Curvature Compensation

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Chi-Wa, U.; Liu, Cong; Martins, Rui P.; Lam, Chi-Seng

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Floating-Domain Integrated GaN Driver Techniques for DC-DC Converters: A Review

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Mu, Xuchu; Zhao, Guangshu; Zhao, Anyang; Jiang, Yang; Law, Man-Kay; Takamiya, Makoto; Mak, Pui-In; Martins, Rui P.

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2023

A High-Current Scalable Parallel LDO Scheme With Analog-Digital Merged Control for Small Current-Sharing Mismatch

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

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2023

A 10.8-to-37.4 Gb/s Reference-Less FD-Less Single-Loop Quarter-Rate Bang-Bang Clock and Data Recovery Employing Deliberate-Current-Mismatch Wide-Frequency-Acquisition Technique

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Wang, Lin; Chen, Yong; Yang, Chaowei; Zhao, Xiaoteng; Mak, Pui-In; Maloberti, Franco; Martins, Rui P.

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2023

A Hybrid Single-Inductor Bipolar Triple-Output DC-DC Converter With High-Quality Positive Outputs for AMOLED Displays

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Mao, Fangyu; Lu, Yan; Maloberti, Franco; Martins, Rui P.

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2023

A 28-nm 368-fJ/Cycle, 0.43%/V Supply-Sensitivity, FLL-Based RC Oscillator Featuring Positive-TC-Only Resistors and $\Delta\Sigma$ -Based Trimming

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Huang, Yunbo; Chen, Yong; Yang, Kaiyuan; Croveti, Paolo; Mak, Pui-In; Martins, Rui P.

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2023

Relative Stability Analysis of Multi-Loop Low Dropout Regulators Using a Sub-Loop Superposition Method

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Yang, Zihan; Huang, Mo; Lu, Yan; Martins, Rui P.

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2023

A High-PCE Range-Extension CMOS Rectifier Employing Advanced Topology Amalgamation Technique for Ambient RF Energy Harvesting

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

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2023

A 0.016mm² Active Area 4GHz Fully Ring-Oscillator-Based Cascaded Fractional-N PLL With Burst-Mode Sampling

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Zhong, Junlin; Yang, Xiaofeng; Martins, Rui P.; Zhu, Yan; Chan, Chi-Hang

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2023

A 0.4-V 8400-m² Voltage Reference in 65-nm CMOS Exploiting Well-Proximity Effect

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Che, Chengyu; Lei, Ka-Meng; Martins, Rui P.; Mak, Pui-In

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2023

A 0.5V 22.5ppm/degree celsius Bandgap Voltage Reference With Leakage Current Injection for Curvature Correction

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Lee, Chon-Fai; U, Chi-Wa; Martins, Rui P.; Lam, Chi-Seng

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2023

A 1-A Switching LDO With 40-mV Dropout Voltage and Fast DVS

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Mao, Xiangyu; Lu, Yan; Martins, Rui P.

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2023

Universal Stability Criterion for Type-I Sampling Phase-Locked Loops

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Huang, Yunbo; Chen, Yong; Mak, Pui-In; Martins, Rui P.

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2023

A 27-dBm, 0.92-GHz CMOS Power Amplifier With Mode Switching and a High-Q Compact Inductor (HQCI) Achieving a 30 Back-Off PAE

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Rawat, Arvind Singh; Rajendran, Jagadheswaran; Mariappan, Selvakumar; Kumar, Narendra; Mak, Pui-In; Martins, Rui P.

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2023

An FPGA-Based Transformer Accelerator Using Output Block Stationary Dataflow for Object Recognition Applications

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Zhao, Zhongyu; Cao, Rujian; Un, Ka-Fai; Yu, Wei-Han; Mak, Pui-In; Martins, Rui P.

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A Reconfigurable CMOS Stack Rectifier With 22.8-dB Dynamic Range Achieving 47.91% Peak PCE for IoT/WSN Application

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Churchill, Kishore Kumar Pakkirisami; Ramiah, Harikrishnan; Choo, Alexander; Chong, Gabriel; Chen, Yong; Mak, Pui-In; Martins, Rui P.

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2023

A High-Performance Dual-Topology CMOS Rectifier With 19.5-dB Power Dynamic Range for RF-Based Hybrid Energy Harvesting

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Choo, Alexander; Ramiah, Harikrishnan; Churchill, Kishore Kumar Pakkirisami; Chen, Yong; Mekhilef, Saad; Mak, Pui-In; Martins, Rui P.

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Sub-5-Minute Ultrafast PCR using Digital Microfluidics

BIOSENSORS & BIOELECTRONICS

Wan, Liang; Li, Mingzhong; Law, Man-Kay; Mak, Pui-In; Martins, Rui P.; Jia, Yanwei

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2023

A CMOS Hall Sensors Array With Integrated Readout Circuit Resilient to Local Magnetic Interference From Current-Carrying Traces

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Zou, Hengchen; Lei, Ka-Meng; Martins, Rui P.; Mak, Pui-In

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2023

An Ultra-Low-Voltage Single-Crystal Oscillator-Timer (XO-Timer) Delivering 16-MHz and 32.258-kHz Clocks for Sub-0.5-V Energy-Harvesting BLE Radios in 28-nm CMOS

IEEE OPEN JOURNAL OF CIRCUITS AND SYSTEMS

Lin, Liwen; Lei, Ka-Meng; Mak, Pui-In; Martins, Rui P.

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2023

A Multimode CMOS Vision Sensor With On-Chip Motion Direction Detection and Simultaneous Energy Harvesting Capabilities (vol 22, pg 12808, 2022)

IEEE SENSORS JOURNAL

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Wideband Continuous-Time MASH Delta-Sigma Modulators: A Tutorial Review

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Qi, Liang; Liu, Yuekai; Sin, Sai-Weng; Xing, Xinpeng; Wang, Guoxing; Ortmanns, Maurits; Martins, Rui P.

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A Highly Integrated Tri-Path Hybrid Buck Converter With Reduced Inductor Current and Self-Balanced Flying Capacitor Voltage

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

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A Dual-Branch Series-Parallel Hybrid Buck DC-DC Converter With Flying Capacitor Voltage Auto-Balancing

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Wang, Chuang; Lu, Yan; Li, Xiuping; Martins, Rui P.

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A 6-to-7.5-GHz 54-fsrms Jitter Type-II Reference-Sampling PLL Featuring a Gain-Boosting Phase Detector for In-Band Phase-Noise Reduction

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Xu, Tailong; Zhong, Shenke; Yin, Jun; Mak, Pui-In; Martins, Rui P.

A 266- μ W Bluetooth Low-Energy (BLE) Receiver Featuring an N -Path Passive Balun-LNA and a Pipeline Down-Mixing BB-Extraction Scheme Achieving 77-dB SFDR and -3-dBm OOB-B-1dB

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Shao, Haijun; Mak, Pui-In; Qi, Gengzhen; Martins, Rui P.

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A 1.7-3.6 GHz 20 MHz-Bandwidth Channel-Selection N-Path Passive-LNA Using a Switched-Capacitor-Transformer Network Achieving 23.5 dBm OB-IIP3 and 3.4-4.8 dB NF

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Constant-Frequency and Noncommunication-Based Inductive Power Transfer Converter for Battery Charging

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Iam, Io-Wa; Hoi, Lok-U; Huang, Zhicong; Gong, Cheng; Lam, Chi-Seng; Mak, Pui-In; Da Silva Martins, Rui Paulo

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Background Timing Mismatch Calibration Techniques in High-Speed Time-Interleaved ADCs: A Tutorial Review

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Guo, Mingqiang; Sin, Sai-Weng; Qi, Liang; Xu, Dengke; Wang, Guoxing; Martins, Rui P.

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A 20 MHz Bandwidth 79 dB SNDR SAR-Assisted Noise-Shaping Pipeline ADC With Gain and Offset Calibrations

IEEE JOURNAL OF SOLID-STATE CIRCUITS

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Ratiometric fluorescence analysis for miR-141 detection with hairpin DNA-templated silver nanoclusters

JOURNAL OF MATERIALS CHEMISTRY C

Liu, Meiqing; Shen, Ren; Li, Haoran; Jia, Yanwei; Mak, Pui-In; Martins, Rui P.

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Evaluation and Perspective of Analog Low-Dropout Voltage Regulators: A Review

IEEE ACCESS

Chyan, Tan Yee; Ramiah, Harikrishnan; Hatta, S. F. Wan Muhamad; Lai, Nai Shyan; Lim, Chee-Cheow; Chen, Yong; Mak, Pui-In; Martins, Rui P.

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Fully-Integrated Timers for Ultra-Low-Power Internet-of-Things Nodes-Fundamentals and Design Techniques

IEEE ACCESS

Loo, Mikki How-Wen; Ramiah, Harikrishnan; Lei, Ka-Meng; Lim, Chee Cheow; Lai, Nai Shyan; Mak, Pui-In; Martins, Rui P.

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A 108-nW 0.8-mm² Analog Voice Activity Detector Featuring a Time-Domain CNN With Sparsity-Aware Computation and Sparsified Quantization in 28-nm CMOS

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Chen, Feifei; Un, Ka-Fai; Yu, Wei-Han; Mak, Pui-In; Martins, Rui P.

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Arithmetic Progression Switched-Capacitor DC-DC Converter Topology With Soft VCR Transitions and Quasi-Symmetric Two-Phase Charge Delivery

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A Reconfigurable Single-Stage Asymmetrical Full-Wave Step-Down Rectifier for Bidirectional Device-to-Device Wireless Fast Charging

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Mao, Fangyu; Lu, Yan; Martins, Rui P.

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2022

A Sub-0.25-pJ/bit 47.6-to-58.8-Gb/s Reference-Less FD-Less Single-Loop PAM-4 Bang-Bang CDR With a Deliberate-Current-Mismatch Frequency Acquisition Technique in 28-nm CMOS

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Zhao, Xiaoteng; Chen, Yong; Wang, Lin; Mak, Pui-In; Maloberti, Franco; Martins, Rui P.

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A Scalable High-Current High-Accuracy Dual-Loop Four-Phase Switching LDO for Microprocessors

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A 0.0285-mm² 0.68-pJ/bit Single-Loop Full-Rate Bang-Bang CDR Without Reference and Separate FD Pulling Off an 8.2-Gb/s/ μ s Acquisition Speed of the PAM-4 Input in 28-nm CMOS

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Zhao, Xiaoteng; Chen, Yong; Mak, Pui-In; Martins, Rui P.

An Inherent Gain Error Tolerance Noise-Shaping SAR-Assisted Pipeline ADC With Code-Counter-Based Offset Calibration

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A 3.3-GS/s 6-b Fully Dynamic Pipelined ADC With Linearized Dynamic Amplifier

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Switched-Capacitor Bandgap Voltage Reference for IoT Applications

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Miniaturized Energy Harvesting Systems Using Switched-Capacitor DC-DC Converters

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

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A Fully-Integrated Ambient RF Energy Harvesting System with 423- μ W Output Power

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A 3-Phase Resonant Switched-Capacitor Converter for Data Center 48-V Rack Power Distribution

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A Single-Stage Dual-Output Regulating Rectifier With Hysteretic Current-Wave Modulation

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A Two-Phase Three-Level Buck Converter With Cross-Connected Flying Capacitors for Inductor Current Balancing

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A 0.003-mm² 440fsRMS-Jitter and-64dBc-Reference-Spur Ring-VCO-Based Type-I PLL Using a Current-Reuse Sampling Phase Detector in 28-nm CMOS

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A 0.14-to-0.29-pJ/bit 14-GBaud/s Trimodal (NRZ/PAM-4/PAM-8) Half-Rate Bang-Bang Clock and Data Recovery (BBCDR) Circuit in 28-nm CMOS

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A 1.7-to-2.7GHz 35-38% PAE Multiband CMOS Power Amplifier Employing a Digitally-Assisted Analog Pre-Distorter (DAAPD) Reconfigurable Linearization Technique

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An FPGA-Based Energy-Efficient Reconfigurable Convolutional Neural Network Accelerator for Object Recognition Applications

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A Wide-PCE-Dynamic-Range CMOS Cross-Coupled Differential-Drive Rectifier for Ambient RF Energy Harvesting

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Chong, Gabriel; Ramiah, Harikrishnan; Yin, Jun; Rajendran, Jagadheswaran; Mak, Pui-In; Martins, Rui P.

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Cost-Effective Compensation Design for Output Customization and Efficiency Optimization in Series/Series-Parallel Inductive Power Transfer Converter

IEEE TRANSACTIONS ON INDUSTRIAL ELECTRONICS

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An NMOS Digital LDO With NAND-Based Analog-Assisted Loop in 28-nm CMOS

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A 4- μm Diameter SPAD Using Less-Doped N-Well Guard Ring in Baseline 65-nm CMOS

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A 2.4-GHz Mid-Field CMOS Wireless Power Receiver Achieving 46% Maximum PCE and 163-mW Output Power

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

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IEEE JOURNAL OF SOLID-STATE CIRCUITS

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IEEE ACCESS

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A 5 GS/s 29 mW Interleaved SAR ADC With 48.5 dB SNDR Using Digital-Mixing Background Timing-Skew Calibration for Direct Sampling Applications

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An 8-Bit 10-GS/s 16x Interpolation-Based Time-Domain ADC With <1.5-ps Uncalibrated Quantization Steps

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IEEE JOURNAL OF SOLID-STATE CIRCUITS

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A 1.6-GS/s 12.2-mW Seven-/Eight-Way Split Time-Interleaved SAR ADC Achieving 54.2-dB SNDR With Digital Background Timing Mismatch Calibration

IEEE JOURNAL OF SOLID-STATE CIRCUITS

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A 76.6-dB-SNDR 50-MHz-BW 29.2-mW Multi-Bit CT Sturdy MASH With DAC Non-Linearity Tolerance

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A 0.096-mm² 1-20-GHz Triple-Path Noise-Canceling Common-Gate Common-Source LNA With Dual Complementary pMOS-nMOS Configuration

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Yu, Haohong; Chen, Yong; Boon, Chirn Chye; Mak, Pui-In; Martins, Rui P.

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A 3.3-mW 25.2-to-29.4-GHz Current-Reuse VCO Using a Single-Turn Multi-Tap Inductor and Differential-Only Switched-Capacitor Arrays With a 187.6-dBc/Hz FOM

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Digital Battery Management Unit With Built-In Resistance Compensation, Modulated Frequency Detection and Multi-Mode Protection for Fast, Efficient and Safe Charging

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Turning on/off satellite droplet ejection for flexible sample delivery on digital microfluidics

LAB ON A CHIP

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2020

Clip-to-release on amplification (CRoA): a novel DNA amplification enhancer on and off microfluidics

LAB ON A CHIP

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Cell-based drug screening on microfluidics

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Hydrodynamic-flow-enhanced rapid mixer for isothermal DNA hybridization kinetics analysis on digital microfluidics platform

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IEEE JOURNAL OF SOLID-STATE CIRCUITS

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A 0.2-V Energy-Harvesting BLE Transmitter With a Micropower Manager Achieving 25% System Efficiency at 0-dBm Output and 5.2-nW Sleep Power in 28-nm CMOS

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Yang, Shiheng; Yin, Jun; Yi, Haidong; Yu, Wei-Han; Mak, Pui-In; Martins, Rui P.

A 550- μ W 20-kHz BW 100.8-dB SNDR Linear-Exponential Multi-Bit Incremental $\Sigma\Delta$ ADC With 256 Clock Cycles in 65-nm CMOS

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Wang, Biao; Sin, Sai-Weng; Seng-Pan, U.; Maloberti, Franco; Martins, Rui P.

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2019

A 0.0056-mm²-249-dB-FoM All-Digital MDLL Using a Block-Sharing Offset-Free Frequency-Tracking Loop and Dual Multiplexed-Ring VCOs

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Yang, Shiheng; Yin, Jun; Mak, Pui-In; Martins, Rui P.

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2019

A 40-Gb/s PAM-4 Transmitter Using a 0.16-pJ/bit SST-CML-Hybrid (SCH) Output Driver and a Hybrid-Path 3-Tap FFE Scheme in 28-nm CMOS

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Fan, Chao; Yu, Wei-Han; Mak, Pui-In; Martins, Rui P.

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A 0.0071-mm² 10.8pspp-Jitter 4 to 10-Gb/s 5-Tap Current-Mode Transmitter Using a Hybrid Delay Line for Sub-1-UI Fractional De-Emphasis

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Chen, Yong; Mak, Pui-In; Yang, Zunsong; Boon, Chirn Chye; Martins, Rui P.

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2019

A 0.12-mm² 1.2-to-2.4-mW 1.3-to-2.65-GHz Fractional-N Bang-Bang Digital PLL With 8- μ s Settling Time for Multi-ISM-Band ULP Radios

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A 0.0018-mm² 153% Locking-Range CML-Based Divider-by-2 With Tunable Self-Resonant Frequency Using an Auxiliary Negative-gm Cell

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CMOS Cross-Coupled Differential-Drive Rectifier in Subthreshold Operation for Ambient RF Energy Harvesting-Model and Analysis

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Chong, Gabriel; Ramiah, Harikrishnan; Yin, Jun; Rajendran, Jagadheswaran; Wong, Wei Ru; Mak, Pui-In; Martins, Rui P.

被引用次数: 24

2019

Fully Integrated High Voltage Pulse Driver Using Switched-Capacitor Voltage Multiplier and Synchronous Charge Compensation in 65-nm CMOS

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Wu, Jiangchao; Lei, Ka-Chon; Leong, Hou-Man; Jiang, Yang; Law, Man-Kay; Mak, Pui-In; Martins, Rui P.

被引用次数: 10

2019

A Comparative Study of 8-Phase Feedforward-Coupling Ring VCOs

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Sun, Iat-Fai; Yin, Jun; Mak, Pui-In; Martins, Rui P.

被引用次数: 14

2019

A 0.5-V 0.4-to-1.6-GHz 8-Phase Bootstrap Ring-VCO Using Inherent Non-Overlapping Clocks Achieving a 162.2-dBc/Hz FoM

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Jiang, Tongquan; Yin, Jun; Mak, Pui-In; Martins, Rui P.

被引用次数: 12

2019

A 5.1-to-7.3 mW, 2.4-to-5 GHz Class-C Mode-Switching Single-Ended-Complementary VCO Achieving > 190 dBc/Hz FoM

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Lim, Chee Cheow; Ramiah, Harikrishnan; Yin, Jun; Kumar, Narendra; Mak, Pui-In; Martins, Rui P.

被引用次数: 11

2019

A 0.044-mm² 0.5-to-7-GHz Resistor-Plus-Source-Follower-Feedback Noise-Cancelling LNA Achieving a Flat NF of 3.3±0.45 dB

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Yu, Haohong; Chen, Yong; Boon, Chirn Chye; Li, Chenyang; Mak, Pui-In; Martins, Rui P.

被引用次数: 64

2019

Micro- and nanofabrication NMR technologies for point-of-care medical applications - A review

MICROELECTRONIC ENGINEERING

Dupre, Antoine; Lei, Ka-Meng; Mak, Pui-In; Martins, Rui P.; Peng, Weng Kung

被引用次数: 35

2019

An Analog-Assisted Tri-Loop Digital Low-Dropout Regulator

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Huang, Mo; Lu, Yan; Seng-Pan, U.; Martins, Rui P.

被引用次数: 95

2018

Algorithmic Voltage-Feed-In Topology for Fully Integrated Fine-Grained Rational Buck-Boost Switched-Capacitor DC-DC Converters

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Jiang, Yang; Law, Man-Kay; Mak, Pui-In; Martins, Rui P.

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2018

An Inverse-Class-F CMOS Oscillator With Intrinsic-High-Q First Harmonic and Second Harmonic Resonances

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lim, Chee Cheow; Ramiah, Harikrishnan; Yin, Jun; Mak, Pui-In; Martins, Rui P.

被引用次数: 52

2018

Low-Phase-Noise Wideband Mode-Switching Quad-Core-Coupled mm-wave VCO Using a Single-Center-Tapped Switched Inductor

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Peng, Yatao; Yin, Jun; Mak, Pui-In; Martins, Rui P.

被引用次数: 55

2018

A Regulation-Free Sub-0.5-V 16-/24-MHz Crystal Oscillator With 14.2-nJ Startup Energy and 31.8- μ W Steady-State Power

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lei, Ka-Meng; Mak, Pui-In; Law, Man-Kay; Martins, Rui P.

被引用次数: 25

2018

A 0.18-V 382- μ W Bluetooth Low-Energy Receiver Front-End With 1.33-nW Sleep Power for Energy-Harvesting Applications in 28-nm CMOS

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Yi, Haidong; Yu, Wei-Han; Mak, Pui-In; Yin, Jun; Martins, Rui P.

被引用次数: 56

2018

A SAW-Less Tunable RF Front End for FDD and IBFD Combining an Electrical-Balance Duplexer and a Switched-LC N-Path LNA

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Qi, Gengzhen; van Liempd, Barend; Mak, Pui-In; Martins, Rui P.; Craninckx, Jan

被引用次数: 76

2018

Missing-Code-Occurrence Probability Calibration Technique for DAC Nonlinearity With Supply and Reference Circuit Analysis in a SAR ADC

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Wang, Guancheng; Li, Cheng; Zhu, Yan; Zhong, Jianyu; Lu, Yan; Chan, Chi-Hang; Martins, Rui P.

被引用次数: 5

2018

A 0.19 mm² 10 b 2.3 GS/s 12-Way Time-Interleaved Pipelined-SAR ADC in 65-nm CMOS

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Zhu, Yan; Chan, Chi-Hang; Zheng, Zi-Hao; Li, Cheng; Zhong, Jian-Yu; Martins, Rui P.

被引用次数: 14

2018

Nano-Ampere Low-Dropout Regulator Designs for IoT Devices

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Huang, Yuanqing; Lu, Yan; Maloberti, Franco; Martins, Rui P.

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2018

A Wideband Inductorless dB-L near Automatic Gain Control Amplifier Using a Single-Branch Negative Exponential Generator for Wireline Applications

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Kong, Lingshan; Chen, Yong; Boon, Chirn Chye; Mak, Pui-In; Martins, Rui P.

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2018

A Hardware-Efficient Feedback Polynomial Topology for DPD Linearization of Power Amplifiers: Theory and FPGA Validation

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Cheang, Chak-Fong; Mak, Pui-In; Martins, Rui P.

被引用次数: 14

2018

A 36-Gb/s 1.3-mW/Gb/s Duobinary-Signal Transmitter Exploiting Power-Efficient Cross-Quadrature Clocking Multiplexers With Maximized Timing Margin

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Chen, Yong; Mak, Pui-In; Boon, Chirn Chye; Martins, Rui P.

被引用次数: 27

2018

Analysis of Common-Mode Interference and Jitter of Clock Receiver Circuits With Improved Topology

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Yang, Xiaofeng; Zhu, Yan; Chan, Chi-Hang; Seng-Pan, U.; Martins, Rui P.

被引用次数: 8

2018

A 0.7-2.5 GHz, 61% EIRP System Efficiency, Four-Element MIMO TX System Exploiting Integrated Power-Relaxed Power Amplifiers and an Analog Spatial De-Interleaver

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Yu, Wei-Han; Un, Ka-Fai; Mak, Pui-In; Martins, Rui P.

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2018

A Coin-Battery-Powered LDO-Free 2.4-GHz Bluetooth Low-Energy Transmitter With 34.7% Peak System Efficiency

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Peng, Xingqiang; Yin, Jun; Yu, Wei-Han; Mak, Pui-In; Martins, Rui P.

被引用次数: 10

2018

A 0.032-mm² 0.15-V Three-Stage Charge-Pump Scheme Using a Differential Bootstrapped Ring-VCO for Energy-Harvesting Applications

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Yi, Haidong; Yin, Jun; Mak, Pui-In; Martins, Rui P.

被引用次数: 57

2018

Gain Error Calibrations for Two-Step ADCs: Optimizations Either in Accuracy or Chip Area

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Wang, Guan Cheng; Zhu, Yan; Chan, Chi-Hang; Seng-Pan, U.; Martins, Rui P.

被引用次数: 2

2018

Nano-Watt Class Energy-Efficient Capacitive Sensor Interface With On-Chip Temperature Drift Compensation

IEEE SENSORS JOURNAL

Zhang, Tan-Tan; Law, Man-Kay; Mak, Pui-In; Vai, Mang-I; Martins, Rui P.

被引用次数: 17

2018

Fully Integrated Inductor-Less Flipping-Capacitor Rectifier for Piezoelectric Energy Harvesting

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Chen, Zhiyuan; Law, Man-Kay; Mak, Pui-In; Ki, Wing-Hung; Martins, Rui P.

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2017

A 0.038-mm² SAW-Less Multiband Transceiver Using an N-Path SC Gain Loop

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Qi, Gengzhen; Mak, Pui-In; Martins, Rui P.

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2017

A 2.4-GHz ZigBee Transmitter Using a Function-Reuse Class-F DCO-PA and an ADPLL Achieving 22.6% (14.5%) System Efficiency at 6-dBm (0-dBm) P_{out}

IEEE JOURNAL OF SOLID-STATE CIRCUITS

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2017

A Handheld High-Sensitivity Micro-NMR CMOS Platform With B-Field Stabilization for Multi-Type Biological/Chemical Assays

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lei, Ka-Meng; Heidari, Hadi; Mak, Pui-In; Law, Man-Kay; Maloberti, Franco; Martins, Rui P.

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2017

A Dual-Output Wireless Power Transfer System With Active Rectifier and Three-Level Operation

IEEE TRANSACTIONS ON POWER ELECTRONICS

Lu, Yan; Huang, Mo; Cheng, Lin; Ki, Wing-Hung; Seng-Pan, U.; Martins, Rui P.

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2017

An Area-Efficient and Tunable Bandwidth-Extension Technique for a Wideband CMOS Amplifier Handling 50+Gb/s Signaling

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Chen, Yong; Mak, Pui-In; Yu, Haohong; Boon, Chirn Chye; Martins, Rui P.

被引用次数: 31

2017

A 0.35-V 520- μ W 2.4-GHz Current-Bleeding Mixer With Inductive-Gate and Forward-Body Bias, Achieving > 13-dB Conversion Gain and > 55-dB Port-to-Port Isolation

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Tan, Gim Heng; Ramiah, Harikrishnan; Mak, Pui-In; Martins, Rui P.

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2017

A High-Voltage-Enabled Class-D Polar PA Using Interactive AM-AM Modulation, Dynamic Matching, and Power-Gating for Average PAE Enhancement

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Yu, Wei-Han; Peng, Xingqiang; Mak, Pui-In; Martins, Rui P.

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2017

A 7.8-mW 5-b 5-GS/s Dual-Edges-Triggered Time-Based Flash ADC

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Chan, Chi-Hang; Zhu, Yan; Sin, Sai-Weng; Seng-Pan, U.; Martins, Rui P.; Maloberti, Franco

被引用次数: 28

2017

A 73.9%-Efficiency CMOS Rectifier Using a Lower DC Feeding (LDCF) Self-Body-Biasing Technique for Far-Field RF Energy-Harvesting Systems

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Moghaddam, Amin Khalili; Chuah, Joon Huang; Ramiah, Harikrishnan; Ahmadian, Jalil; Mak, Pui-In; Martins, Rui P.

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2017

A 3D microblade structure for precise and parallel droplet splitting on digital microfluidic chips

LAB ON A CHIP

Dong, Cheng; Jia, Yanwei; Gao, Jie; Chen, Tianlan; Mak, Pui-In; Vai, Mang-I; Martins, Rui P.

被引用次数: 35

2017

Metastability in SAR ADCs

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Chan, Chi-Hang; Zhu, Yan; Sin, Sai-Weng; Murmann, Boris; Seng-Pan, U.; Martins, R. P.

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2017

A Wide Input Range Dual-Path CMOS Rectifier for RF Energy Harvesting

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Lu, Yan; Dai, Haojuan; Huang, Mo; Law, Man-Kay; Sin, Sai-Weng; U, Seng-Pan; Martins, Rui P.

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2017

A 27-Gb/s Time-Interleaved Duobinary Transmitter Achieving 1.44-mW/Gb/s FOM in 65-nm CMOS

IEEE MICROWAVE AND WIRELESS COMPONENTS LETTERS

Chen, Yong; Mak, Pui-In; Boon, Chirn Chye; Martins, Rui P.

被引用次数: 16

2017

A 0.45 V 147-375 nW ECG Compression Processor With Wavelet Shrinkage and Adaptive Temporal Decimation Architectures

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

leong, Chio-In; Li, Mingzhong; Law, Man-Kay; Mak, Pui-In; Vai, Mang I.; Martins, Rui P.

Active-Passive $\Delta \Sigma$ Modulator for High-Resolution and Low-Power Applications

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Hussain, Arshad; Sin, Sai-Weng; Chan, Chi-Hang; U, Seng-Pan (Ben); Maloberti, Franco; Martins, Rui P.

被引用次数: 23

2017

A Single-Chip Solar Energy Harvesting IC Using Integrated Photodiodes for Biomedical Implant Applications

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Chen, Zhiyuan; Law, Man-Kay; Mak, Pui-In; Martins, Rui P.

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2017

A digital microfluidic system for loop-mediated isothermal amplification and sequence specific pathogen detection

SCIENTIFIC REPORTS

Wan, Liang; Chen, Tianlan; Gao, Jie; Dong, Cheng; Wong, Ada Hang-Heng; Jia, Yanwei; Mak, Pui-In; Deng, Chu-Xia; Martins, Rui P.

被引用次数: 64

2017

Drug screening of cancer cell lines and human primary tumors using droplet microfluidics

SCIENTIFIC REPORTS

Wong, Ada Hang-Heng; Li, Haoran; Jia, Yanwei; Mak, Pui-In; da Silva Martins, Rui Paulo; Liu, Yan; Vong, Chi Man; Wong, Hang Cheong; Wong, Pak Kin; Wang, Haitao; Sun, Heng; Deng, Chu-Xia

被引用次数: 70

2017

A Time-Interleaved Ring-VCO with Reduced $1/f^3$ Phase Noise Corner, Extended Tuning Range and Inherent Divided Output

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Yin, Jun; Mak, Pui-In; Maloberti, Franco; Martins, Rui P.

被引用次数: 28

2016

A μ NMR CMOS Transceiver Using a Butterfly-Coil Input for Integration With a Digital Microfluidic Device Inside a Portable Magnet

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lei, Ka-Meng; Mak, Pui-In; Law, Man-Kay; Martins, Rui P.

被引用次数: 22

2016

A 2- μ m InGaP/GaAs Class-J Power Amplifier for Multi-Band LTE Achieving 35.8-dB Gain, 40.5% to 55.8% PAE and 28-dBm Linear Output Power

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Jagadheswaran, U. R.; Ramiah, Harikrishnan; Mak, Pui-In; Martins, Rui P.

被引用次数: 26

2016

Sub-7-second genotyping of single-nucleotide polymorphism by high-resolution melting curve analysis on a thermal digital microfluidic device

LAB ON A CHIP

Chen, Tianlan; Jia, Yanwei; Dong, Cheng; Gao, Jie; Mak, Pui-In; Martins, Rui P.

被引用次数: 31

2016

CMOS biosensors for in vitro diagnosis - transducing mechanisms and applications

LAB ON A CHIP

Lei, Ka-Meng; Mak, Pui-In; Law, Man-Kay; Martins, Rui P.

被引用次数: 39

2016

Uniform Quantization Theory-Based Linearity Calibration for Split Capacitive DAC in an SAR ADC

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Liu, Jianwei; Zhu, Yan; Chan, Chi-Hang; Sin, Sai-Weng; Seng-Pan, U.; Martins, Rui Paulo da Silva

被引用次数: 14

2016

A 1.1 μ W CMOS Smart Temperature Sensor With an Inaccuracy of ± 0.2 $^{\circ}$ C (3σ) for Clinical Temperature Monitoring

IEEE SENSORS JOURNAL

Law, Man-Kay; Lu, Sanfeng; Wu, Tao; Bermak, Amine; Mak, Pui-In; Martins, Rui P.

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2016

A palm-size μ NMR relaxometer using a digital microfluidic device and a semiconductor transceiver for chemical/biological diagnosis

ANALYST

Lei, Ka-Meng; Mak, Pui-In; Law, Man-Kay; Martins, Rui P.

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2015

Nested-Current-Mirror Rail-to-Rail-Output Single-Stage Amplifier With Enhancements of DC Gain, GBW and Slew Rate

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Yan, Zushu; Mak, Pui-In; Law, Man-Kay; Martins, Rui P.; Maloberti, Franco

被引用次数: 66

2015

A 0.02 mm² 59.2 dB SFDR 4th-Order SC LPF With 0.5-to-10 MHz Bandwidth Scalability Exploiting a Recycling SC-Buffer Biquad (vol 50, pg 1988, 2015)

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Mohan, P. V. Ananda; Zhao, Yaohua; Mak, Pui-In; Martins, Rui P.; Maloberti, Franco

被引用次数: 0

2015

A 0.02 mm² 59.2 dB SFDR 4th-Order SC LPF With 0.5-to-10 MHz Bandwidth Scalability Exploiting a Recycling SC-Buffer Biquad

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Zhao, Yaohua; Mak, Pui-In; Martins, Rui P.; Maloberti, Franco

被引用次数: 6

2015

Self-Reconfiguration Property of a Mixed Signal Controller for Improving Power Quality Compensation During Light Loading

IEEE TRANSACTIONS ON POWER ELECTRONICS

Wong, Man-Chung; Yang, Yan-Zheng; Lam, Chi-Seng; Choi, Wai-Hei; Dai, Ning Yi; Wu, Yajie; Wong, Chi-Kong; Sin, Sai-Weng; Chio, U-Fat; U, Seng-Pan; Martins, Rui P.

被引用次数: 9

2015

A Sub-GHz Wireless Transmitter Utilizing a Multi-Class-Linearized PA and Time-Domain Wideband-Auto I/Q-LOFT Calibration for IEEE 802.11af WLAN

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Un, Ka-Fai; Yu, Wei-Han; Cheang, Chak-Fong; Qi, Gengzhen; Mak, Pui-In; Martins, Rui P.

被引用次数: 9

2015

A Combinatorial Impairment-Compensation Digital Predistorter for a Sub-GHz IEEE 802.11af-WLAN CMOS Transmitter Covering a 10x-Wide RF Bandwidth

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Cheang, Chak-Fong; Un, Ka-Fai; Yu, Wei-Han; Mak, Pui-In; Martins, Rui P.

被引用次数: 10

2015

Wideband Receivers: Design Challenges, Tradeoffs and State-of-the-Art

IEEE CIRCUITS AND SYSTEMS MAGAZINE

Lin, Fujian; Mak, Pui-In; Martins, Rui P.

被引用次数: 36

2015

A 0.07 mm² 2.2 mW 10 GHz Current-Reuse Class-B/C Hybrid VCO Achieving 196-dBc/Hz FoMA

IEEE MICROWAVE AND WIRELESS COMPONENTS LETTERS

Amin, Md. Tawfiq; Yin, Jun; Mak, Pui-In; Martins, Rui P.

被引用次数: 22

2015

Improving the Linearity and Power Efficiency of Active Switched-Capacitor Filters in a Compact Die Area

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Zhao, Yaohua; Mak, Pui-In; Law, Man-Kay; Martins, Rui P.

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2015

On the droplet velocity and electrode lifetime of digital microfluidics: voltage actuation techniques and comparison

MICROFLUIDICS AND NANOFUIDICS

Dong, Cheng; Chen, Tianlan; Gao, Jie; Jia, Yanwei; Mak, Pui-In; Vai, Mang-I; Martins, Rui P.

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2015

NMR-DMF: a modular nuclear magnetic resonance-digital microfluidics system for biological assays

ANALYST

Lei, Ka-Meng; Mak, Pui-In; Law, Man-Kay; Martins, Rui P.

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2014

A Sub-GHz Multi-ISM-Band ZigBee Receiver Using Function-Reuse and Gain-Boosted N-Path Techniques for IoT Applications

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lin, Zhicheng; Mak, Pui-In; Martins, Rui P.

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2014

An RF-to-BB-Current-Reuse Wideband Receiver With Parallel N-Path Active/Passive Mixers and a Single-MOS Pole-Zero LPF

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lin, Fujian; Mak, Pui-In; Martins, Rui P.

被引用次数: 40

2014

A 2.4 GHz ZigBee Receiver Exploiting an RF-to-BB-Current-Reuse Mixer plus Hybrid Filter Topology in 65 nm CMOS

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lin, Zhicheng; Mak, Pui-In; Martins, Rui P.

被引用次数: 45

2014

A 0.14-mm² 1.4-mW 59.4-dB-SFDR 2.4-GHz ZigBee/WPAN Receiver Exploiting a Split-LNTA+50% LO Topology in 65-nm CMOS

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Lin, Zhicheng; Mak, Pui-In; Martins, Rui P.

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2014

A Sine-LO Square-Law Harmonic-Rejection Mixer-Theory, Implementation, and Application

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

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Analysis and Modeling of a Gain-Boosted N-Path Switched-Capacitor Bandpass Filter

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Lin, Zhicheng; Mak, Pui-In; Martins, Rui P.

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2014

A 0.137 mm² 9 GHz Hybrid Class-B/C QVCO With Output Buffering in 65 nm CMOS

IEEE MICROWAVE AND WIRELESS COMPONENTS LETTERS

Amin, Md. Tawfiq; Mak, Pui-In; Martins, Rui P.

被引用次数: 5

2014

Enhancing the performances of recycling folded cascode OpAmp in nanoscale CMOS through voltage supply doubling and design for reliability

INTERNATIONAL JOURNAL OF CIRCUIT THEORY AND APPLICATIONS

Mak, Pui-In; Liu, Miao; Zhao, Yaohua; Martins, Rui P.

被引用次数: 4

2014

An intelligent digital microfluidic system with fuzzy-enhanced feedback for multi-droplet manipulation

LAB ON A CHIP

Gao, Jie; Liu, Xianming; Chen, Tianlan; Mak, Pui-In; Du, Yuguang; Vai, Mang-I; Lin, Bingcheng; Martins, Rui P.

被引用次数: 61

2013

A 53-to-75-mW, 59.3-dB HRR, TV-Band White-Space Transmitter Using a Low-Frequency Reference LO in 65-nm CMOS

IEEE JOURNAL OF SOLID-STATE CIRCUITS

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被引用次数: 11

2013

A 0.016 mm² 144- μ W Three-Stage Amplifier Capable of Driving 1-to-15 nF Capacitive Load With >0.95-MHz GBW (vol 48, pg 527, 2013)

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Yan, Zushu; Mak, Pui-In; Law, Man-Kay; Martins, R. P.

被引用次数: 0

2013

A 0.016-mm² 144- μ W Three-Stage Amplifier Capable of Driving 1-to-15 nF Capacitive Load With >0.95-MHz GBW

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Yan, Zushu; Mak, Pui-In; Law, Man-Kay; Martins, Rui P.

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2013

Construction of a microfluidic chip, using dried-down reagents, for LATE-PCR amplification and detection of single-stranded DNA

LAB ON A CHIP

Jia, Yanwei; Mak, Pui-In; Massey, Conner; Martins, Rui P.; Wangh, Lawrence J.

被引用次数: 18

2013

15-nW Biopotential LPFs in 0.35- μ m CMOS Using Subthreshold-Source-Follower Biquads With and Without Gain Compensation

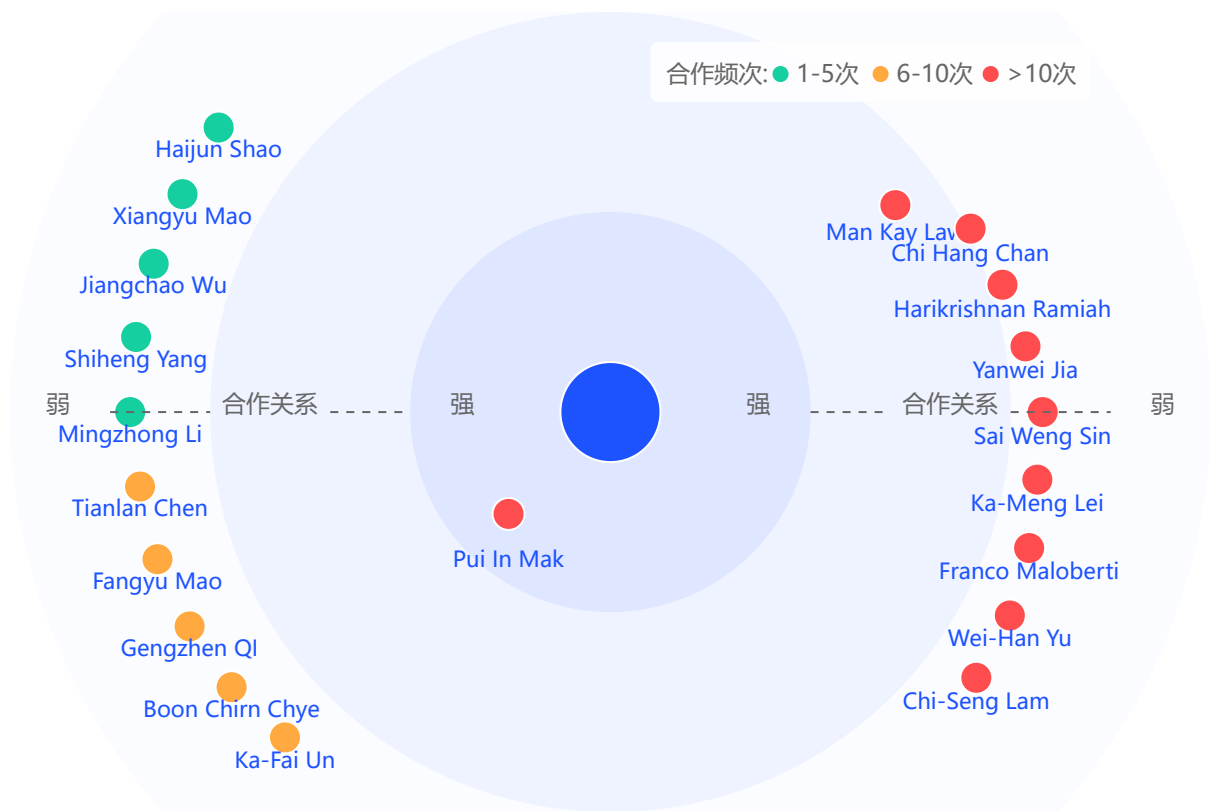
IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Zhang, Tan-Tan; Mak, Pui-In; Vai, Mang-I; Mak, Peng-Un; Law, Man-Kay; Pun, Sio-Hang; Wan, Feng; Martins, Rui P.

被引用次数: 54

2013

学术合作



Pui In Mak

合作次数: 161

澳门大学

Man Kay Law

合作次数: 37

澳门大学

Chi Hang Chan

合作次数: 24

澳门大学

Harikrishnan Ramiah

合作次数: 22

马来亚大学

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